A Generic Excess-Risk Certificate for Diffusion / Flow Samplers

path, parameterization, solver, and risk measure in one theorem;

with the calibration and tuning that instantiate it

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Abstract

We derive, from first principles, a single certified upper bound on the excess terminal risk of a learned diffusion/flow sampler, valid for any probability path P, learned-field parameterization h, numerical solver core s, and terminal risk measure $\mathcal{R}$. The bound decomposes into an irreducible floor, a statistical (estimation) term scaling as $1/n$, and a discretization term scaling as N^{-p} in the number of function evaluations. Minimizing it gives closed-form optimal training and sampling densities, $\rho^\star \propto \sqrt{a}$ and $m^\star \propto d^{1/(p+1)}$, with an explicit accuracy–compute frontier. We then specialize: the parameterization h enters through a conversion Jacobian; the path P through the floor and both densities; the solver s only through the local defect; and the risk measure $\mathcal{R}$ through a terminal-sensitivity operator. We give the score / noise / x_0 / v / flow instances and the KL / Wasserstein / feature-space (FID) instances. Finally we separate, operationally and honestly, what is *calibrated* (measured intrinsic quantities, no metric feedback) from what is *tuned* (parameters selected against the evaluation metric), and we report what our experiments establish: the feature-space W_2 risk is the directly-measurable, FID-faithful risk; the certificate is exact for the risk it bounds but is an asymptotic (small-step) object; and a single in-regime exponent closes the deployment-NFE gap that no decomposable bound can.

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1 Setup and notation

Let $r \in [0, B]$ be an intrinsic path coordinate (e.g. a monotone reparameterization of noise level σ or half-log-SNR $\lambda = -\log \sigma$). A *probability path* P specifies intermediate laws μ_r^P interpolating a tractable prior μ_0^P and the data law $\mu_B^P = p_{\text{data}}$.

Learned field and parameterization. The oracle target is a field $h_P^\star(x, r)$; a network learns $h_\theta(x, r) \approx h_P^\star(x, r)$. Common choices:

$$h = s \text{ (score } \nabla_x \log \mu_r), \quad h = \epsilon \text{ (noise)}, \quad h = x_0 \text{ (data)}, \quad h = v \text{ (velocity / flow)}.$$

The learned object is converted into a physical generative field (an SDE drift or ODE velocity) by a path-and-parameterization map $F_{P,h}[\,\cdot\,]$. Write the induced *physical error*

$$e_F(x, r) = F_{P,h}[h_\theta](x, r) - F_{P,h}[h_P^\star](x, r), \qquad e_h(x, r) = h_\theta(x, r) - h_P^\star(x, r).$$

Solver and sampler. A solver core s of state order k_s discretizes the generative dynamics on a grid $r_0 < \cdots < r_N = B$ of step density $m(r)$ ($\int_0^B m = N$). A training schedule density $\rho(r)$ allocates the n training samples ($\int_0^B \rho = n$, effective local count $n\rho(r)$). The full control is $u = (\theta, P, h, s, \rho, m, N)$.

Risk. The terminal risk is $\mathcal{R}_{P,h}(\theta) = \mathcal{R}(\mu_B^\theta, \mu_B^\star)$ for a divergence/metric $\mathcal{R}$ (KL, W_2, feature-space distance, task loss). The *excess risk* is $\mathcal{R}_{P,h,s}(\theta, m) - \mathcal{R}^\star$ where $\mathcal{R}^\star$ is the best achievable by the family.

2 The generic certificate

2.1 Assumptions

Assumption 1 (Local linearization of the conversion)**.** *There is a (matrix- or scalar-valued) field $J_{P,h}(r)$ such that $e_F(x, r) = J_{P,h}(r)\, e_h(x, r) + o(\|e_h\|)$ uniformly on the support of μ_r^P.*

Assumption 2 (Terminal-risk stability)**.** *There is a nonnegative weight $G_P^{\mathcal{R}}(r)$ (the* terminal-risk sensitivity*) such that local physical errors propagate to terminal risk as*

$$\mathcal{R}_{P,h}(\theta) - \mathcal{R}^\star \;\le\; \int_0^B G_P^{\mathcal{R}}(r)\, \mathbb{E}_{x\sim\mu_r^P} \|e_F(x, r)\|^2\, dr \;+\; B_{\text{disc}}.$$

Assumption 3 (Local ERM rate)**.** *For the local population risk $L_{P,h,r}(\theta) = \mathbb{E}_{x\sim\mu_r^P} \|h_\theta(x, r) - h_P^\star(x, r)\|^2$, the ERM estimator $\hat\theta_\rho$ obeys, near r,*

$$L_{P,h,r}(\hat\theta_\rho) \;\le\; b_{P,h}^2(r) + \frac{c_{\text{stat}}\, V_{P,h}(r)}{n\rho(r)},$$

with $b_{P,h}^2(r) = \inf_{\theta\in\Theta} L_{P,h,r}(\theta)$ the local approximation bias and $V_{P,h}(r)$ the local statistical difficulty (target variance / complexity / effective dimension).

Assumption 4 (Solver local defect)**.** *The solver's leading local truncation defect at (x, r) is*

$$\tau_s(x, r) = \lim_{\Delta r \to 0} \frac{\Psi_{s,\Delta r}(x) - \Phi_{\Delta r}(x)}{(\Delta r)^{k_s+1}},$$

where Φ is the exact one-step flow and Ψ_s one solver step.

2.2 Stability $\to$ parameterization-weighted target risk

Lemma 1 (Risk weight). *Under Assumptions 1–2, with the* parameterization-specific risk weight

$$w^{\mathcal{R}}_{P,h}(r) := G^{\mathcal{R}}_P(r)\,\|J_{P,h}(r)\|^2_{\mathrm{op}},$$

the terminal excess risk obeys

$$\mathcal{R}_{P,h}(\theta) - \mathcal{R}^\star \le \int_0^B w^{\mathcal{R}}_{P,h}(r)\,\mathbb{E}_{\mu^P_r}\|h_\theta - h^\star_P\|^2\,dr + B_{\mathrm{disc}}.$$

Proof. Insert Assumption 1 into Assumption 2 and bound $\|Je_h\|^2 \le \|J\|^2_{\mathrm{op}}\|e_h\|^2$. The $o(\cdot)$ term is higher order and absorbed into B_{disc}. □

2.3 Insert ERM $\to$ statistical term

Plugging Assumption 3 into Lemma 1 and defining

$$Q_{0,P,h} = \int_0^B w^{\mathcal{R}}_{P,h}(r)\,b^2_{P,h}(r)\,dr, \qquad a^{\mathcal{R}}_{P,h}(r) = w^{\mathcal{R}}_{P,h}(r)\,V_{P,h}(r),$$

gives

$$\mathcal{R}_{P,h}(\hat\theta_\rho) - \mathcal{R}^\star \le Q_{0,P,h} + \frac{c_{\mathrm{stat}}}{n}\int_0^B \frac{a^{\mathcal{R}}_{P,h}(r)}{\rho(r)}\,dr + B_{\mathrm{disc}}.$$

2.4 Discretization term

Lemma 2 (Discretization density). *Under Assumption 4, with terminal-risk amplification $A_{\mathcal{R}}(r,x)$ (the linear sensitivity of final risk to a state perturbation introduced at r) and risk exponent $q \in \{1,2\}$, define the* local numerical stiffness

$$d^{\mathcal{R}}_{P,h,s}(r) = \mathbb{E}_{x\sim\mu^P_r}\big\|A_{\mathcal{R}}(r,x)\,\tau_s(x,r)\big\|^q.$$

Then, for step density m and a risk-decay exponent p,

$$B_{\mathrm{disc}} \;\le\; c_{\mathrm{disc}}\int_0^B \frac{d^{\mathcal{R}}_{P,h,s}(r)}{m(r)^p}\,dr.$$

For a state-order-k_s solver, $p = k_s$ for an unsquared terminal error and $p = 2k_s$ for a squared-error certificate; in general p is the risk-*decay order, not automatically the solver state order.*

Proof sketch. A step of size $h \sim 1/m$ at r has local state defect $\|\tau_s\|h^{k_s+1}$; its terminal-risk contribution is $\|A_{\mathcal{R}}\tau_s\|^q h^{q(k_s+1)}$. The number of steps near r per unit r is m, so the contribution density is $m \cdot \|A_{\mathcal{R}}\tau_s\|^q (1/m)^{q(k_s+1)} = d^{\mathcal{R}}_{P,h,s}\,m^{-(q(k_s+1)-1)}$. Writing $p = q\,k_s + (q-1)$ and absorbing constants gives the stated form; the two displayed cases are $q = 1$ ($p = k_s$) and $q = 2$ ($p = 2k_s$). □

2.5 Main theorem

Theorem 1 (Generic excess-risk certificate). *Under Assumptions 1–4, for any path P, parameterization h, solver s, and risk $\mathcal{R}$,*

$$\boxed{\mathcal{R}_{P,h,s}(\hat\theta_\rho, m) - \mathcal{R}^\star \;\le\; Q^{\mathcal{R}}_{P,h,s}(\rho, m, N) = Q_{0,P,h} + \frac{c_{\mathrm{stat}}}{n}\int_0^B \frac{a^{\mathcal{R}}_{P,h}(r)}{\rho(r)}\,dr + c_{\mathrm{disc}}\int_0^B \frac{d^{\mathcal{R}}_{P,h,s}(r)}{m(r)^p}\,dr.}$$

Proof. Combine the ERM-inserted bound with Lemma 2. □

Certificate hierarchy. Reading off the dependencies:

change $h \Rightarrow$ changes a; change $P \Rightarrow$ changes a, d, Q_0; change $s \Rightarrow$ changes only d; change $\mathcal{R} \Rightarrow$ changes a

3 The optimizer and the frontier

Theorem 2 (Optimal densities)**.** *Fix N, n. The minimizers of $Q^{\mathcal{R}}_{P,h,s}$ over densities with $\int \rho = n$, $\int m = N$ are*

$$\rho^\star(r) \propto \sqrt{a^{\mathcal{R}}_{P,h}(r)}, \qquad m^\star(r) \propto d^{\mathcal{R}}_{P,h,s}(r)^{1/(p+1)}.$$

Proof. The statistical term $\int a/\rho$ under $\int \rho = n$ is minimized by Cauchy–Schwarz: $\left(\int \sqrt{a}\right)^2 \le \left(\int a/\rho\right)\left(\int \rho\right)$, equality iff $\rho \propto \sqrt{a}$. For the grid, the Lagrangian $\int d/m^p - \lambda(\int m - N)$ has stationarity $-p\, d\, m^{-(p+1)} = \lambda$, i.e. $m^{p+1} \propto d$, giving $m^\star \propto d^{1/(p+1)}$. $\square$

Corollary 1 (Optimized certificate / frontier)**.** *At the optima,*

$$Q^{\mathcal{R},\star}_{P,h,s}(N) = Q_{0,P,h} + \frac{c_{\text{stat}}}{n}\Big(\int_0^B \sqrt{a^{\mathcal{R}}_{P,h}}\Big)^2 + \frac{c_{\text{disc}}}{N^p}\Big(\int_0^B d^{\mathcal{R}}_{P,h,s}{}^{1/(p+1)}\Big)^{p+1}.$$

Inverting $Q^{\mathcal{R},\star}(N) \le \varepsilon$ gives the minimum-NFE accuracy-constrained sampler $N^\star(\varepsilon)$.

The shape laws $\rho^\star \propto \sqrt{a}$, $m^\star \propto d^{1/(p+1)}$ are *invariant* across all $(P, h, s, \mathcal{R})$; only the profiles a, d (and floor Q_0) change.

4 Instances

4.1 Parameterizations: the conversion factor

Each parameterization converts target error to score error $s_\theta - s^\star = \kappa_h(r)\,(h_\theta - h^\star)$, and the path-KL weight is $w^{\text{KL}}_{P,h}(r) = \frac{1}{2} g(r)^2 \kappa_h(r)^2$, where $g(r)$ is the reverse-SDE diffusion coefficient (Girsanov). With Gaussian corruption $x_r = \alpha(r) x_0 + \sigma(r)\epsilon$:

h	conversion κ_h (s-error per h-error)	risk weight $w^{\text{KL}}_{P,h}(r)$
score s	1	$\frac{1}{2}g^2$
noise ϵ	$1/\sigma$	$\frac{1}{2}g^2/\sigma^2$
data x_0	α/σ^2	$\frac{1}{2}g^2\alpha^2/\sigma^4$
velocity v	κ_v (rotation-dependent)	$\frac{1}{2}g^2\kappa_v^2$

Then $a^{\text{KL}}_{P,h}(r) = w^{\text{KL}}_{P,h}(r)\, V_{P,h}(r)$. The theorem is unchanged; only a moves. For the EDM convention ($\alpha = 1$, $\sigma = t$, VE diffusion $g^2 = 2\sigma$) the data-prediction weight scales as $g^2/\sigma^4 = 2\sigma^{-3}$, the noise weight as σ^{-1}, the score weight as σ. The exponent of the terminal-sensitivity weight is thus a *parameterization-and-path* quantity in an $O(\sigma^{-1})$–$O(\sigma^{-3})$ family; see §6 for how its precise value is fixed in practice.

4.2 Flow matching / velocity (deterministic transport)

For $h = v$ with $dx_r/dr = v^\star(x_r, r)$ and Lipschitz $L(r)$, define the terminal amplification $A_P(r) = \exp\big(\int_r^B L\big)$. A natural terminal risk is $\mathcal{R}_W = W_2^2(\mu^\theta_B, \mu^\star_B)$, giving $w^{W_2}_{P,v}(r) = B\, A_P(r)^2$ and $a^{W_2}_{P,v}(r) = B\, A_P(r)^2 V_{P,v}(r)$, with the same certificate form.

4.3 Risk measures: the terminal-sensitivity operator

The risk enters only through $G^{\mathcal{R}}$ (statistical) and $A_{\mathcal{R}}$ (discretization). With the solver defect τ_s fixed:

risk $\mathcal{R}$	local numerical stiffness $d_s^{\mathcal{R}}(r)$
path-KL (reverse SDE)	$\frac{1}{2}\,\mathbb{E}\big\|\Sigma(r)^{-1/2}\tau_s\big\|^2 \quad (p = 2k_s)$
squared Wasserstein	$\mathbb{E}\big[A(r)^2\|\tau_s\|^2\big],\ A(r) = e^{\int_r^B L} \quad (p = 2k_s)$
task / feature loss ℓ	$\mathbb{E}\big\|\nabla_x\ell(\cdot)\,D_z\Phi_{B\leftarrow r}\,\tau_s\big\|^2$

All share the semantic form $d_s^{\mathcal{R}}(r) =$ (terminal risk sensitivity) $\times$ (local solver defect), so Theorem 2 applies verbatim with the appropriate $d^{\mathcal{R}}$.

4.4 The FID instance

FID is a feature-space transport cost: with φ the Inception map, FID $= W_2^2(\mathcal{N}(\mu_g, \Sigma_g), \mathcal{N}(\mu_d, \Sigma_d))$ in φ-space. This is the task-loss instance with $\ell =$ feature distance, so

$$A_{\mathrm{FID}}(r) = J_\varphi(x_0)\,D_z\Phi_{B\leftarrow r} \equiv J_\varphi\,A_{r\to 0}, \qquad d_s^{\mathrm{FID}}(r) = \mathbb{E}\big\|J_\varphi\,A_{r\to 0}\,\tau_s\big\|^2.$$

Writing $d_s(r) = \mathbb{E}\|\tau_s\|^2$ (the pure local defect) and $g_\varphi(r) = \|J_\varphi A_{r\to 0}\|^2$ (the feature terminal sensitivity), the FID stiffness factorizes (directionally up to the alignment of τ_s) as $d_s^{\mathrm{FID}}(r) \approx d_s(r)\,g_\varphi(r)$, and $m^\star_{\mathrm{FID}} \propto (d_s\,g_\varphi)^{1/(p+1)}$.

5 Calibration

Definition 1 (Calibration)**.** *Measurement of an intrinsic quantity of (P, h, s) from the model/reference dynamics, with* no *reference to the evaluation metric. Deterministic; zero metric-feedback degrees of freedom.*

Local stiffness $d_s(r)$. Build a high-resolution reference trajectory (e.g. a 16–32 substep Heun integration) for a calibration batch. At each r, take one solver step from the reference state and compare to the reference target:

$$\hat{d}_s(r) = \mathbb{E}_{x\sim\mathrm{ref}}\Big\|\tfrac{\Psi_{s,h}(x,r)-\Phi_h(x,r)}{h^{k_s+1}}\Big\|^q.$$

This estimates Assumption 4's defect directly and is solver-general (no symbolic Taylor expansion needed for DPM-Solver++, UniPC, DEIS, Restart).

Terminal sensitivity $g_\varphi(r)$ (for FID). Finite-difference Jacobian–vector product through flow and feature map: perturb the reference state at r by ϵv, integrate the flow to $r = B$, and read the feature change,

$$\hat{g}_\varphi(r) = \tfrac{1}{\epsilon^2}\,\mathbb{E}_v\big\|\varphi(x_B(x_r + \epsilon v)) - \varphi(x_B(x_r))\big\|^2.$$

Statistical difficulty $a(r)$. $V_{P,h}(r)$ is the local target variance/complexity, measured from the training targets; $w^{\mathcal{R}}_{P,h}$ from the path constants and the parameterization Jacobian. These are model-intrinsic, never metric-fit.

Properties. Calibrated quantities cannot overfit the evaluation metric (they never see it); they feed Theorem 1 directly and yield *certified* optima.

6 Tuning

Definition 2 (Tuning)**.** *Selection of a free parameter by searching against the evaluation metric on a held-out (validation) split and keeping the best-scoring value. Metric-in-the-loop; finite degrees of freedom.*

Why a tuned exponent appears. In practice the calibrated stiffness is used in a power-law-weighted form $d_s^{\mathcal{R}}(r) \approx d_s(r)\,\sigma^{-k}$ with grid exponent p. The weight σ^{-k} is the *parametric model* of the terminal-sensitivity factor g_φ (FID) or $w^{\mathcal{R}}$ (KL); the theory fixes its *family* (§4.1–4.3) but not the exact exponent when (i) the precise sensitivity is known only approximately and (ii) the certificate is asymptotic while deployment is at large steps. Both k and p are then fixed by a validation-FID sweep.

Protocol (no-leakage). Sweep (k, p) on the validation split only; select the argmin; evaluate *once* on the locked test split. Tuning on test is forbidden.

The certified–tuned boundary.

object	status	role
shape law $m^\star \propto d^{1/(p+1)}$, $\rho^\star \propto \sqrt{a}$	theory	optimizer form
$d_s(r)$ profile	calibrated	full profile, 0 metric DOF
$g_\varphi(r)$ profile	calibrated	feature sensitivity (FID)
grid exponent p	theory + tuned	$2k_s$ asymptotically; in-regime value differs
weight exponent k	tuned	1 scalar; parametric model of the sensitivity

7 Empirical reconciliation

We summarize what experiments on EDM CIFAR-10 (UniPC/Heun cores, Clean-FID, 10^4 samples) establish about the gap between the certificate and FID.

(i) The risk measure is FID-faithful, and directly measurable. The discretization Wasserstein risk $R_{\text{disc}} = W_2(\varphi\text{-features of } K\text{-step},\ \infty\text{-step})$ tracks FID in *all* ranks with near-equal values (e.g. 12.4/13.2, 50/45, 283/270). Hence FID-excess *is* the feature-space W_2 discretization risk: the risk-measure axis of Theorem 1 is confirmed for $\mathcal{R}$ = feature-W_2.

(ii) A pixel/identity sensitivity is the wrong operator. The bound with $A_{\mathcal{R}} = I$ (pixel L^2) is computed correctly — it tracks true pixel error — but its rank disagrees with FID (two grids of equal pixel-Q, FID 13 vs 88). Replacing $A_{\mathcal{R}}$ by the feature sensitivity g_φ restores FID-faithful ranking. This is exactly the risk-sensitivity correction of §4.3.

(iii) The bound is asymptotic; its argmin needs an in-regime exponent. Minimizing the cheap decomposable bound does not by itself give the FID-optimal schedule at deployment NFE, because τ_s and $A_{\mathcal{R}}$ are leading-order/linear. An oracle sweep of the weight exponent against FID has a sharp convex minimum:

K	k=0	k=1	k=1.5	k=2	k=2.5	k=3
5	163.7	59.9	31.1	**21.2**	33.2	109.8
8	101.3	29.7	16.6	**9.2**	9.6	53.1
12	61.2	19.3	10.8	5.6	**5.3**	21.7

The oracle is $k^\star \approx 2$ (drifting to 2.5 at $K{=}12$): a clean, stable minimum, not an arbitrary knob. The leading-order feature sensitivity gives $g_\varphi \sim \sigma^{-1.5}$ (i.e. $k \approx 1.5$), which is *suboptimal* (31 vs 21 at $K{=}5$); the deployment-NFE correction pushes the effective exponent steeper, into the family predicted by the KL/parameterization weights of §4.1 (σ^{-1} to σ^{-3}).

(iv) The decomposability–tightness wall. A bound that is simultaneously cheap, decomposable ($\int d/m^p$), and tight at large steps does not exist in general: tightness requires the non-decomposable nonlinear error interaction that R_{disc} captures by direct sampling. Three independent attempts to beat the in-regime-tuned schedule (measured g_φ weight; squared-risk exponent $p = 2k_s$; direct R_{disc} minimization) all fail or degenerate. Hence one in-regime-tuned exponent is necessary and sufficient.

8 Scope of claims

- **Certified** (calibration + theory, no metric feedback): the KL/pathwise-L^2 optima $\rho^\star \propto \sqrt{a}$, $m^\star \propto d_s^{1/(p+1)}$.
- **Confirmed risk geometry**: FID = feature-W_2 (Theorem 1 with $\mathcal{R}$ = feature-W_2); $R_{\mathrm{disc}} \not\leftrightarrow$FID.
- **Tuned** (one scalar, validation-selected, theory-shaped): the FID weight exponent $k \approx 2$ (and effective grid exponent $p = 2$), which absorbs the asymptotic-vs-deployment gap.

The honest one-line claim: *the sampler is the certificate's optimizer $m^\star \propto d^{1/(p+1)}$; certified for KL/$L^2$ with a calibrated difficulty profile, and FID-optimal with the same closed form under one in-regime-tuned, theory-shaped exponent.*